of a modest mountain) should have evaporated completely away by now, and black holes a few times heavier than this should still be evaporating strongly. Such black holes have horizons about the size of an atomic nucleus.

A large portion of the energy emitted in the evaporation of such holes should now be in the form of gamma rays (high-energy photons) traveling randomly through the Universe. Such gamma rays do exist, but in amounts and with properties that are readily explained in other ways. The absence of excess gamma rays tells us (according to calculations by Hawking and Page) that there now are no more than about 300 tiny, strongly evaporating black holes in each cubic light-year of space; and this, in turn, tells us that matter in the big bang cannot have had an extremely soft equation of state.

Skeptics will argue that the absence of excess gamma rays might have another interpretation: Perhaps many small black holes were formed in the big bang, but we physicists understand quantum fields in curved spacetime far less well than we think we do, and thus we are misleading ourselves when we believe that black holes evaporate. I and my colleagues resist such skepticism because of the seeming perfection with which the standard laws of curved spacetime and the standard laws of quantum fields mesh to give us a nearly *unique* set of laws for quantum fields in curved spacetime. Nevertheless, we would feel rather more comfortable if astronomers could find observational evidence of black-hole evaporation.

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1. It might seem counterintuitive that Hawking's area-increase theorem permits any of the holes' mass at all to be emitted as gravitational waves. Readers comfortable with algebra may find satisfaction in the example of two nonspinning holes that coalesce to produce a single, larger nonspinning hole. The surface area of a nonspinning hole is proportional to the square of its horizon circumference, which in turn is proportional to the square of the hole's mass. Thus, Hawking's theorem insists that the sum of the squares of the initial holes' masses must be less than the square of the final hole's mass. A little algebra shows that this constraint on the masses permits the final hole's mass to be less than the sum of the initial holes' masses, and thus permits some of the initial masses to be emitted as gravitational waves.

2. A more precise definition of the apparent horizon is given in Box 12.1 below.

3. The laws of quantum mechanics guarantee that the number of ways to distribute the atoms and molecules is always finite, and never infinite. In defining the entropy, physicists often multiply the logarithm of this number of ways by a constant that will be irrelevant to us, $\log_e 10 \times k$, where $\log_e 10$ is the "natural logarithm" of 10, that is, 2.30258 . . . , and k is "Boltzmann's constant," 1.38062×10^{-16} erg per degree Celsius. Throughout this book I shall ignore this constant.